

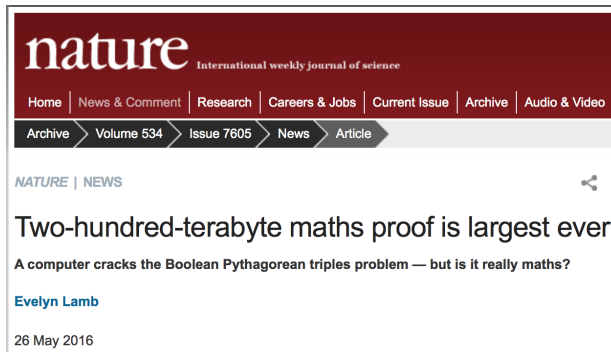
Der DPLL Algorithmus

Prof. Dr. Christoph Benzmüller
Freie Universität Berlin

Adressierte Themen:

- ▶ Wissensrepräsentation und Schließen
- ▶ Propositionale Logik und SAT Solving
- ▶ Davis-Putnam-Logemann-Loveland (DPLL) Algorithmus

Beispiel einer Erfolgsgeschichte



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nature International weekly journal of science

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NATURE | NEWS 

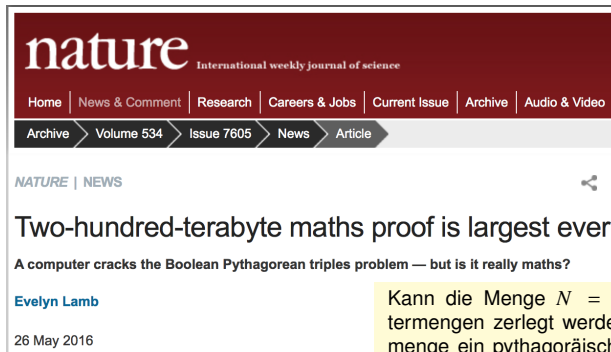
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[Evelyn Lamb](#)

26 May 2016

Beitrag von
M. Heule
O. Kullmann
V. Marek



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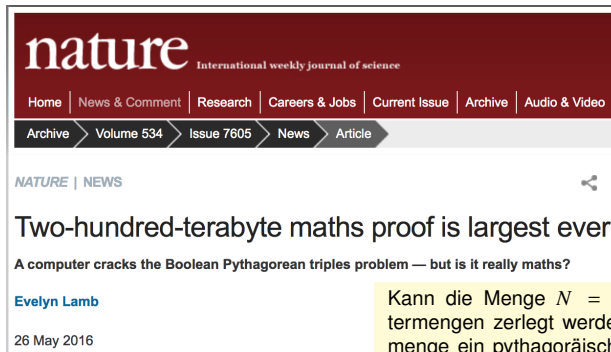
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Kann die Menge $N = \{1, 2, \dots, n\}$ in zwei Untermengen zerlegt werden, so dass keine Untermenge ein pythagoräisches Triple (a, b, c) enthält mit $a^2 + b^2 = c^2$?

$n = 10$

$$3^2 + 4^2 = 5^2$$

(wähle die Farbe der anderen Zahlen beliebig)



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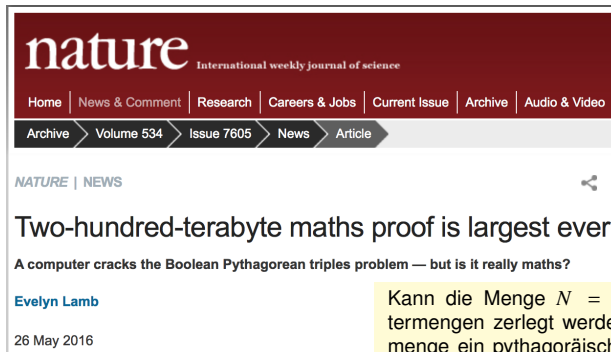
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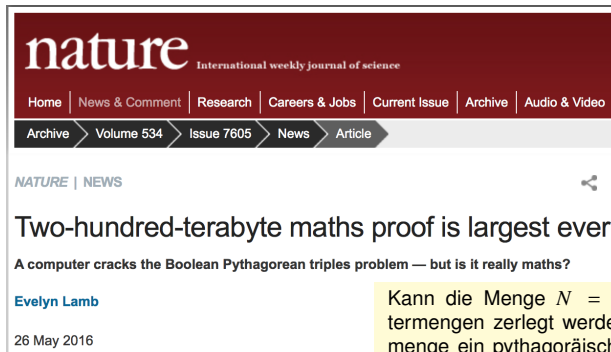
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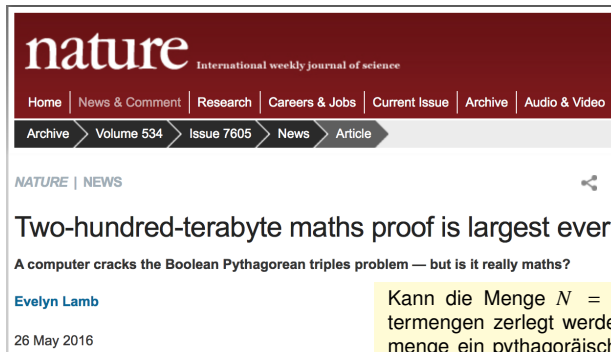
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Verboten: nur eine Farbe!



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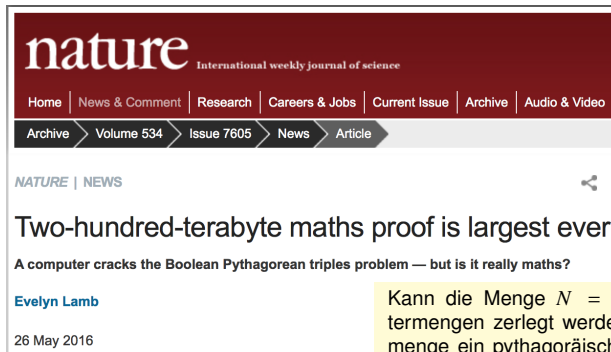
$n = 20$

$$3^2 + 4^2 = 5^2$$

$$5^2 + 12^2 = 13^2$$

$$8^2 + 15^2 = 17^2$$

(wähle die Farbe der anderen Zahlen beliebig)



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$n = 30$

$$3^2 + 4^2 = 5^2$$

$$5^2 + 12^2 = 13^2$$

$$8^2 + 15^2 = 17^2$$

$$7^2 + 24^2 = 25^2$$

(choose color of the other numbers arbitrarily)



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$n = 40$

$$3^2 + 4^2 = 5^2$$

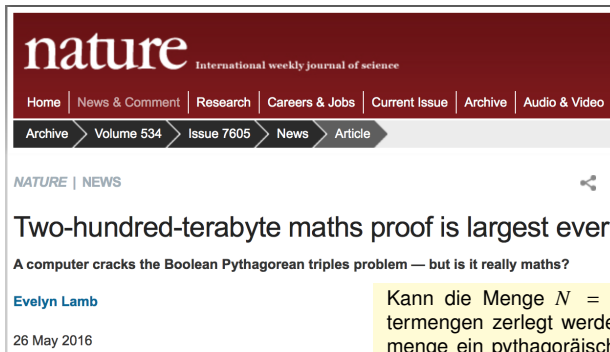
$$5^2 + 12^2 = 13^2$$

$$8^2 + 15^2 = 17^2$$

$$7^2 + 24^2 = 25^2$$

$$20^2 + 21^2 = 29^2$$

$$12^2 + 35^2 = 37^2$$



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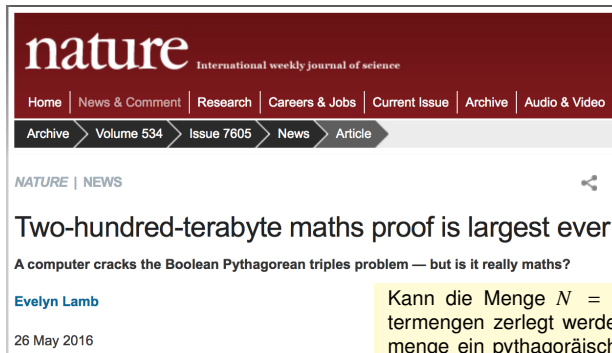
$$8^2 + 15^2 = 17^2$$

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Shown by SAT-Solver:

For $n \geq 7825$ consistent bi-coloring becomes impossible.

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$n = 40$

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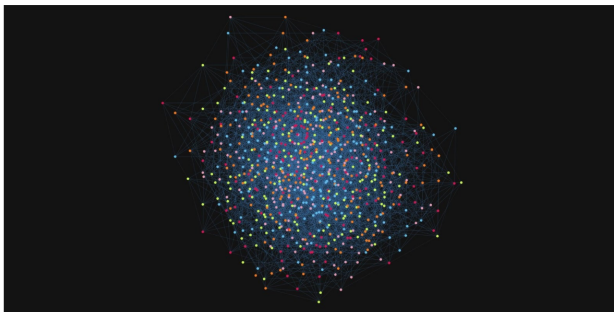
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The Science of Brute Force, M.J.H. Heule, O. Kullmann, Communications of the ACM, Vol. 60 No. 8, Pages 70-79, 2017, DOI:10.1145/3107239

<https://cacm.acm.org/magazines/2017/8/219606-the-science-of-brute-force/fulltext>

EVELYN LAMB SCIENCE 04.30.18 09:00 AM

AN ANTI-AGING PUNDIT SOLVES A DECADES-OLD MATH PROBLEM



This 826-vertex graph requires at least five colors to ensure that no two connected vertices are the same shade. (Click [here](#) for a high-resolution version.)

 OLENA SHMAHALO/QUANTA MAGAZINE; SOURCE: [MARIJN HEULE](#)

Übergeordnetes Thema: Erfüllbarkeitsproblem (SAT) der Aussagenlogik

- ▶ **P vs. NP** (1 Million Dollar Frage, Clay Mathematics Institute)
- ▶ **SAT** ist **NP-vollständig**
- ▶ **SAT-Solver** trotzdem sehr erfolgreich in der Praxis

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Syntax Aussagenlogik

$$s, t ::= \top \mid \perp \mid A \mid \neg s \mid s \vee t \mid s \wedge t \mid s \rightarrow t \mid s \leftrightarrow t$$

Semantik Aussagenlogik: Abbildung nach T (true) oder F (false)

s	t	$\top$	$\perp$	$\neg s$	$s \vee t$	$s \wedge t$	$s \rightarrow t$	$s \leftrightarrow t$
T	T	T	F	F	T	T	T	T
T	F	T	F	F	T	F	F	F
F	T	T	F	T	T	F	T	F
F	F	T	F	T	F	F	T	T

Klauselnormalform bzw. Konjunktive Normalform (CNF)

Beispielformel:

$$(A \wedge \neg B) \leftrightarrow (A \vee B)$$

Beispielformel in CNF:

$$(A \vee \neg B) \wedge (\neg A \vee \neg B) \wedge \neg B$$

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Notation: Konjunktion von Klausel, Klauseln, Literale

CNF

$$(A \vee \neg B) \wedge (\neg A \vee \neg B) \wedge \neg B$$

Notation für Klauseln

$$A\bar{B} \wedge \bar{A}\bar{B} \wedge \bar{B}$$

Notation für Klausel-Listen

$$A\bar{B}, \bar{A}\bar{B}, \bar{B}$$

Annahme (für Rest der Vorlesung)

Darstellung von Klauseln und Klausel-Listen: **Kommutativ und Assoziativ**

Belegungen $[\psi]$ am Beispiel:

 $[A\bar{B}]$ repräsentiert die **Belegung** $\{A \rightarrow \text{T}, B \rightarrow \text{F}\}$ $[\bar{B}]$ ist eine **partielle Belegung** für $A\bar{B}, \bar{A}\bar{B}, \bar{B}$ $[\psi] \models s$ ist **Notation** für: Belegung $[\psi]$ erfüllt Formel s $[\psi]$ wird dann auch als **Modell für** s bezeichnet

Regelsysteme/Transitionssysteme

Grundkenntnisse werden vorausgesetzt

Notation: Konjunktion von Klausel, Klauseln, Literale

CNF

$$(A \vee \neg B) \wedge (\neg A \vee \neg B) \wedge \neg B$$

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Grundkenntnisse werden vorausgesetzt

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CNF

$$(A \vee \neg B) \wedge (\neg A \vee \neg B) \wedge \neg B$$

Notation für Klauseln

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Belegungen $[\psi]$ am Beispiel:

 $[A\bar{B}]$ repräsentiert die **Belegung** $\{A \rightarrow \text{T}, B \rightarrow \text{F}\}$ $[\bar{B}]$ ist eine **partielle Belegung** für $A\bar{B}, \bar{A}\bar{B}, \bar{B}$ $[\psi] \models s$ ist **Notation** für: Belegung $[\psi]$ erfüllt Formel s $[\psi]$ wird dann auch als **Modell** für s bezeichnet

Regelsysteme/Transitionssysteme

Grundkenntnisse werden vorausgesetzt

Notation: Konjunktion von Klausel, Klauseln, Literale

CNF

$$(A \vee \neg B) \wedge (\neg A \vee \neg B) \wedge \neg B$$

Notation für Klauseln

$$A\bar{B} \wedge \bar{A}\bar{B} \wedge \bar{B}$$

Notation für Klausel-Listen

$$A\bar{B}, \bar{A}\bar{B}, \bar{B}$$

Annahme (für Rest der Vorlesung)

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Grundkenntnisse werden vorausgesetzt

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabeformel	(CNF & Initialisierung)
$\square$	$A\bar{B}C, \bar{B}, C\bar{D}, BCD$	(Schritt 1)
	...	(Schritt 2)
	...	
	...	(Schritt n)
$[\bar{B}C]$	$A\bar{B}C, \bar{B}, C\bar{D}, BCD$	(Solved)

Kriterium für “Solved”?

$[\bar{B}C]$ $\underbrace{A\bar{B}C}_{\checkmark}, \underbrace{\bar{B}}_{\checkmark}, \underbrace{C\bar{D}}_{\checkmark}, \underbrace{BCD}_{\checkmark}$ —alle Klauseln sind erfüllt—

(es gilt dann: $[\bar{B}C] \models \text{Eingabeformel}$)

Solved-Regel

$[\psi] \quad \phi$ (Solved)

Bedingung: $[\psi] \models \phi$

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Bedingung: $[\psi] \models \phi$

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	(CNF & Initialisierung)
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}\bar{F}G}, \bar{C}FG, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	(CNF & Initialisierung)
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}\bar{F}G}, \bar{C}FG, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	(CNF & Initialisierung)
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}\bar{F}G}, \bar{C}FG, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	(CNF & Initialisierung)
$[\]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}\bar{F}G}, \bar{C}FG, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Solved)

Unit-Propagation-Regel:

$[\psi] \quad \phi, \varphi L \quad (\text{Unit-Propagation})$

Bedingung:

$\downarrow \quad [\psi] \models \neg\varphi$

$L \notin [\psi]$

$[\psi L] \quad \phi, \varphi L \quad \text{Verarbeite neue Belegungsinformation}$

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	(CNF & Initialisierung)
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
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$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}\bar{F}G}, \bar{C}FG, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	(CNF & Initialisierung)
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}\bar{F}G}, \bar{C}FG, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	(CNF & Initialisierung)
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}\bar{F}G}, \bar{C}FG, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	(CNF & Initialisierung)
$[\]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}\bar{F}G}, \bar{C}FG, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Solved)

Pure-Literal-Regel

$[\psi] \quad \phi, \varphi L \quad (\text{Pure-Literal})$

Bedingung:

↓

L nur mit einheitlicher Polarität in $\phi, \varphi L$

L undefiniert in $[\psi]$

$[\psi L] \quad \phi, \varphi L$

Verarbeite neue Belegungsinformation

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}\bar{F}G}, \bar{C}FG, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	(CNF & Initialisierung)
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, \cancel{C}\bar{G}F, \bar{C}\bar{F}G, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}}FG, \cancel{\bar{C}}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}}\bar{F}G, \bar{D}E, \cancel{\bar{F}}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}}\bar{F}G, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, \cancel{C}\bar{G}F, \bar{C}\bar{F}G, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}}FG, \cancel{\bar{C}}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}}\bar{F}G, \bar{D}E, \cancel{\bar{F}}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}}\bar{F}G, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, \cancel{C}\bar{G}F, \bar{C}\bar{F}G, \bar{C}\bar{F}\bar{G}, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}}FG, \cancel{\bar{C}}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}}\bar{F}G, \bar{D}E, \cancel{\bar{F}}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}}\bar{F}G, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
		(Solved)

Split-Regel:

$[\psi] \quad \phi \quad (\text{Split})$

Bedingung:

$\downarrow$

L und $\bar{L}$ kommen in ϕ vor

L undefiniert in $[\psi]$

$[\psi L_b] \quad \phi$

Verarbeite neue Belegungsinformation

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}\bar{F}G}, \bar{C}FG, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{F}G, \cancel{\bar{F}}G, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{F}}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Split)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{F}}G, \cancel{F}G, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
		(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{F}G, \cancel{\bar{F}}G, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{F}}G, \bar{D}E, \cancel{\bar{F}}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{F}}G, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
		(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Split)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
		(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Split)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
		(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Split)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}\bar{F}G}, \bar{C}FG, \bar{D}E, \bar{F}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
		(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Split)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
		(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, \cancel{BCF}, \cancel{C\bar{G}F}, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, \cancel{BCF}, \cancel{C\bar{G}F}, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
		(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Split)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
		(Solved)

Backtrack Rule:

$[\psi L_b \psi'] \quad \phi, \varphi \quad (\text{Backtrack})$

Bedingung:

$\downarrow$

$[\psi L_b \psi'] \models \neg \varphi$

ψ' enthält keine mit “b” markierten Literale

$[\psi \bar{L}] \quad \phi, \varphi$

Verarbeite neue Belegungsinformation

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Split)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
		(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Split)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
		(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Split)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}\bar{F}G}, \bar{C}FG, \bar{D}E, \bar{F}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
		(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Split)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
		(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	(CNF & Initialisierung)
$[\]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}\bar{F}G}, \bar{C}FG, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Solved)

Solved-Regel

$[\psi] \quad \phi \quad (\text{Solved})$

Bedingung: $[\psi] \models \phi$

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Split)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
		(Solved)

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Split)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}\bar{F}G}, \bar{C}FG, \bar{D}E, \bar{F}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
		(Solved)

Fail-Regel:

$[\psi] \quad \phi, \varphi \quad (\text{Split})$

Bedingung:

$\downarrow$

$[\psi] \models \neg\varphi$

ψ enthält keine Backtrack-Literale

$[] \quad \text{Fail}$

DPLL-Algorithmus — Entwicklung am Beispiel

	Eingabe-Formel	
$[]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(CNF & Initialisierung)
$[\bar{B}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
$[\bar{B}\bar{D}]$	$A\bar{B}, \bar{B}, \cancel{BCF}, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Pure-Literal)
$[\bar{B}\bar{D}C_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}FG}, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \bar{F}\bar{G}$	(Split)
$[\bar{B}\bar{D}C_bF_b]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Split)
$[\bar{B}\bar{D}C_bF_bG]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \cancel{\bar{C}\bar{F}G}, \bar{D}E, \cancel{\bar{F}\bar{G}}$	(Unit-Propagation)
$[\bar{B}\bar{D}C_b\bar{F}]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \cancel{\bar{C}\bar{F}G}, \bar{C}FG, \bar{D}E, \bar{F}\bar{G}$	(Backtrack)
$[\bar{B}\bar{D}C_b\bar{F}G]$	$A\bar{B}, \bar{B}, BCF, C\bar{G}F, \bar{C}FG, \bar{C}\bar{F}G, \bar{D}E, \bar{F}\bar{G}$	(Unit-Propagation)
		(Solved)

Abstrakter DPLL Algorithmus (Regelbasierte Darstellung)

Wende die Regeln erschöpfend an mit folgender Priorität:

1. Solved
2. Fail
3. Backtrack
4. Unit-Propagation
5. Pure-Literal
6. Split

DPLL Algorithmus am Beispiel — Kurzes Beispiel, alle Regeln

[]	an , ãn , ãñ , bn	(Pure Literal)
[b]	an , ãn , ãñ , bn	(Split)
[ba]	an , ãn , ãñ , bn	(Unit Propagation)
[ban]	an , ãn , ãñ , bn	(Widerspruch: n&ñ, Backtracking)
[bã]	an , ãn , ãñ , bn	(Unit Propagation)
[bãn]	an , ãn , ãñ , bn	(Solved)

Möglichkeiten zur weiteren Verbesserung bzw. Optimierung

- ▶ Heuristiken für Split: z.B. wähle das am häufigsten vorkommende Literal (Vorheriges Beispiel: wähle F zuerst – diese Wahl verhindert Backtracking und generiert eine alternative Belegung (Übungsaufgabe))
- ▶ Nicht-chronologisches Backtracking
- ▶ Klausel-Lernen beim Backtracking
- ▶ Effiziente Datenstrukturen
- ▶ Indexing-Techniken, Watchlist
- ▶ ...

Korrektheit und Vollständigkeit: für abstraktes System in Vorlesung

Systeme (z.B. zChaff, MiniSat, PicoSat, Limmat, Lingeling)

Anwendungsbeispiele

Satisfiability modulo Theories